

Rank Ranking Preservation under Bounded Score Perturbations

Research Architecture Runtime
`operator/rank`

2026-07-25

Abstract

Rank is certified via pairwise ranking preservation when all gaps exceed 2ε . Lean: `LEAN_FULL`.

1 Theorems

Theorem 1 (Preservation). *All pairwise gaps $> 2\varepsilon$ preserve pairwise order under $\|\delta\|_\infty \leq \varepsilon$.*

Theorem 2 (Sharpness). *A pairwise gap $\leq 2\varepsilon$ admits a collision adversary.*

2 Lean / certificate

`Research.Operators.Rank.Preservation`.

3 Limitations

Mathlib $\mathbb{R}$ (`REAL_MATHLIB`).